



The Body Acceptance by Others Scale-2: An examination of its psychometric properties in a French-Canadian adult sample¹

Christophe Maïano^{a,b,*}, Viren Swami^{c,d}, Tracy L. Tylka^e, Annie Aimé^a

^a Cyberpsychology Laboratory and Department of Psychoeducation and Psychology, Université du Québec en Outaouais (UQO|Campus de Saint-Jérôme), Saint-Jérôme, Canada

^b Substantive-Methodological Synergy Research Laboratory, Department of Psychology, Concordia University, Montreal, Canada

^c School of Psychology, Sport, and Sensory Sciences, Anglia Ruskin University, Cambridge, United Kingdom

^d Centre for Psychological Medicine, Perdana University, Kuala Lumpur, Malaysia

^e Department of Psychology, The Ohio State University, Columbus, OH, USA

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ABSTRACT

The 13-item Body Acceptance by Others Scale-2 (BAOS-2) is an instrument designed to measure generalised perceptions of body acceptance by others. Here, we examined the psychometric properties of a novel French translation of the BAOS-2. A total of 763 French-speaking adults from Canada were asked to complete the BAOS-2, alongside additional validated measures of positive body image, psychological well-being, and eating patterns. Exploratory factor analysis in a first split-half subsample ($n = 382$) indicated that a unidimensional model with all 13 items had good fit to the data. This model was cross-validated in a second split-half subsample ($n = 381$) using confirmatory factor analysis. A partial factors-only model indicated, as a function of subsamples, that the French BAOS-2 has differential item functioning based on respondent age, body mass index (BMI), and gender identity (men, women). In addition, higher respondent BMI was associated with lower body acceptance in both subsamples. Our results also supported convergent and concurrent validity of the French BAOS-2, with body acceptance by others associated with greater body appreciation and psychological well-being, as well as lower symptoms of maladaptive eating patterns. The present results indicate that the French BAOS-2 has strong psychometric properties in French-speaking Canadian adults.

1. Introduction

Body acceptance by others refers to “a sense that one’s body and its physical characteristics are valued, respected, and unconditionally accepted by important others... irrespective of what others or social domains actually offer at any particular time” (Swami, Todd, Stieger et al., 2021, p. 239). Defined in this way and consistent with the Acceptance Model of Intuitive Eating (Avalos & Tylka, 2006), body acceptance by others is not itself a facet of body image, but is instead theorised as an antecedent of facets of positive body image. More specifically, when individuals have access to external relational connections and social systems that are validating, they are likely to feel empowered to resist self-objectification, disengage from restrictive appearance-related pressures, and adopt adaptive body-related attitudes

that result in more positive body image (Avalos & Tylka, 2006; Tigge-mann, 2019). Indeed, there is a large body of evidence showing that greater body acceptance by others is significantly associated with more positive body image (e.g., Chan et al., 2023; Longhurst & Swami, 2023; Swami et al., 2018, 2023, 2025) and embodied well-being (Moelle & Papps, 2025) in diverse communities.

The construct of body acceptance by others was originally measured using the Body Acceptance by Others Scale (BAOS; Avalos & Tylka, 2006). This is a 10-item instrument in which respondents are asked to rate the perceived acceptance of their body shape and weight, and the degree to which they receive messages that their body shape and weight are “fine”, from five sources (i.e., friends, family, dating partners, mass media, and society). In their initial assessment, Avalos and Tylka (2006) reported that scores on the BAOS had adequate composite reliability,

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* Correspondence to: Université du Québec en Outaouais, Campus de Saint-Jérôme, Département de Psychoéducation et de Psychologie, 5 rue Saint-Joseph, Saint-Jérôme, Québec J7Z 0B7, Canada.

E-mail address: christophe.maiano@uqo.ca (C. Maïano).

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test-retest reliability up to three weeks, and construct validity. However, more recent work has reported difficulties replicating a unidimensional model of BAOS scores (Swami, Furnham, Horne, & Stieger, 2020; Swami, Todd et al., 2020). Additional concerns with the BAOS include its specific focus on body weight and shape rather than body and its physical characteristics more holistically (i.e., what is being measured), the limiting of body acceptance by others to the receipt of messages as opposed to perceived body acceptance (i.e., how the construct is defined), and the focus on specific targets rather than generalised perceptions of body acceptance (i.e., how the construct is measured). Such concerns led to the development of a revised version of the BAOS.

The second iteration of the BAOS – the BAOS-2 (Swami, Todd, Stieger et al., 2021) – is a 13-item instrument that assesses generalised perceptions of body acceptance by others. In adults from the United Kingdom, Swami, Todd, Stieger et al. (2021) reported that scores on the BAOS-2 had a unidimensional factor structure, with adequate composite reliability, test-retest reliability up to four weeks, and good convergent, construct, criterion, discriminant, and incremental validity. Additionally, these authors also reported that the BAOS-2 achieved strong invariance across gender identity and that there was no significant difference in body acceptance by others between women and men. Finally, Swami, Todd, Stieger et al. (2021) also reported that scores on the BAOS-2 achieved strong invariance across adults from the United Kingdom and the United States. Although participants in the United Kingdom were found to have significantly higher (manifest) BAOS-2 scores compared to their counterparts from the United States, the effect size of the difference was negligible.

A small number of studies have examined the psychometric properties of translated versions of the BAOS-2. In terms of factorial validity, studies using a combination of exploratory factor analysis (EFA) and confirmatory factor analysis (CFA) have reported that the BAOS-2 retains its unidimensional factor structure in adults from Malaysia (Swami et al., 2021), China (Wang et al., 2022), Germany (Behrend et al., 2023), Poland (Modrzejewska et al., 2024), and Italy (Nerini et al., 2024). One study using only EFA has also reported that the unidimensional model of the BAOS-2 is supported in Israeli women (Geller et al., 2024). Where it has been assessed, authors of these translations have also reported that the unidimensional model is supported in women and men separately, and that scores achieve strong invariance across gender identity. Additionally, comparisons of scores across women and men have generally indicated non-significant or negligible gender identity differences in all cases.

In addition, there is some evidence that the BAOS-2 is invariant across racialised identities (Malaysian Malays vs. Chinese) and national groups (Malaysia, the United Kingdom, and the United States; Swami et al., 2021). Beyond factorial validity, test adaptation studies have generally reported that the BAOS-2 has adequate convergent validity (e.g., positive associations with facets of positive body image) and concurrent validity (e.g., positive associations with psychological well-being and negative associations with pressure to internalise appearance ideals; Modrzejewska et al., 2024; Nerini et al., 2024; Swami et al., 2021; Wang et al., 2022). Among Chinese adults, Wang et al. (2022) additionally reported that baseline BAOS-2 scores were related to improvements in body and functionality appreciation six months later, indicative of predictive validity. In short, the available evidence suggests that the BAOS-2 retains its robust psychometric properties across translational contexts.

1.1. The French-Canadian context

To add to this body of translational work, we sought to examine the psychometric properties of a novel French translation of the BAOS-2 in Canadian adults. Recently, the BAOS-2 has been translated into French, but with items altered to specifically refer to body acceptance by one's father and mother, respectively (Lacerte et al., 2025). An EFA revealed that scores reduced to two dimensions, reflective of a factor for each

parent, in French-speaking adolescent girls from Canada. However, this does not necessarily indicate true instrument unidimensionality (i.e., the two dimensions are likely an artefact of item wording) and, to our knowledge, the psychometric properties of the French BAOS-2 in its original form has not been assessed. While adding to the range of languages in which the psychometric properties of the BAOS-2 remains an important task (Swami, Todd, Stieger et al., 2021), assessing the instrument's psychometric properties in French Canadian adults may be interesting because of the uniqueness of this population.

First, the Francophone population remains a minority social identity in the Canadian federation (Wu & Wilkes, 2017). This minority status has been suggested to be a reason for higher levels of social connectedness and group-based self-conceptualisations among Francophone Canadians compared to Anglophone Canadians (e.g., Clift, 1989; Latouche, 2006). Additionally, French Canadian (particularly Québécois) roots in Catholicism and a rejection of eighteenth-century liberal individualism (e.g., Dufort et al., 2022; Richer & Laporte, 1973) may provide additional pressures toward collectivist trends. Such influences may heighten sensitivity to social cues and attendance to the judgements and emotional responses of others (Swami et al., 2021), which in turn may affect the way in which body acceptance by others is experienced. Indeed, relatively higher levels of collectivism in French Canada may mean that individuals are highly attuned to the perceptions of others and that body acceptance by others is integral to the development and shaping of one's self-concept. To be sure, the extent to which these historic trends continue to exert an effect on contemporary Francophone Canada is a matter of some debate (e.g., Wu & Dawson, 2022), but nevertheless, Francophone Canada would appear to be an interesting region in which to assess body acceptance by others.

The contemporary body image literature also suggests that body acceptance by others may be an important factor to consider in Francophone Canada. For example, one study of Québécois adolescents showed that nearly a third had received commentary about their bodies or appearance from others, with the majority of commentary originating from close others (e.g., peers, siblings, parents; Aimé & Maïano, 2014). A more recent study of Québécois adolescents reported that nearly a half had received weight-based commentary, primarily from close friends, which in turn was associated with appearance dissatisfaction (Gagné et al., 2023). However, gaps remain in our understanding of these issues, particularly as body acceptance by others is not necessarily the polar opposite of appearance-related commentary (Swami, Todd, Stieger et al., 2021). In other words, a lack of appearance-related commentary by itself cannot be used to understand or estimate high levels of unconditional body acceptance from others. A fuller understanding of the construct of body acceptance by others and its correlates in the French-Canadian context would, therefore, be highly informative.

1.2. The present study

In the present work, our primary objective was to assess the psychometric properties of a novel French translation of the BAOS-2 in a sample of French-Canadian adults. In terms of assessing factorial validity, current guidelines recommend the use of an EFA-to-CFA strategy when a construct is hypothesised to be unidimensional, as is the case for the BAOS-2 (Swami, Todd, Stieger et al., 2021). This strategy combines an exploratory approach in which the most suitable model for a dataset is extracted, followed by a confirmatory approach in which the best-fitting model is cross-validated in a new sample or subsample (Swami & Barron, 2019; Swami, Maïano, & Morin, 2023). Additionally, we examined invariance of the best factor solution across two independent samples and differential item functioning (DIF) of participant gender identity (women vs. men), age, and body mass index (BMI) as function of the two independent samples. DIF is used as an alternative to the more popular multi-group CFA when characteristics occur as continuous variables (e.g., age and BMI) or when testing for measurement biases as a joint function of multiple variables (Penfield & Lam,

2000).

Additionally, we examined the construct validity of the French BAOS-2. To do so, we examined associations between BAOS-2 scores and scores on additional variables that have been validated for use in French-speaking populations. More specifically, to assess convergent validity, we examined associations with a core facet of positive body image, namely body appreciation, with the expectation of a positive and moderate-to-strong association. To assess concurrent validity, we considered associations with psychological well-being (symptoms of anxiety and depression, life satisfaction, and self-esteem), eating patterns (dieting behaviour and orthorexia), and self-compassion. Positive and moderate associations between body acceptance by others and life satisfaction, self-esteem, self-compassion, and non-pathological (or “healthy”) orthorexia were expected. Moreover, we expected negative and moderate associations with anxiety, depression, dieting behaviours, and orthorexia nervosa.

2. Method

2.1. Participants

The participants of the present study were 763 French-speaking Canadian adults aged between 17 and 90 years old ($M = 50.87$, $SD = 15.87$). Of this sample, 52.9% identified as women, 45.9% as men, 0.7% as non-binary, and 0.5% as another gender. In terms of education, 0.3% had completed elementary education, 14.2% had completed high school education, 15.5% had completed a diploma of vocational studies, 22.6% had completed a pre-university programme (CEGEP), 30.6% had an undergraduate degree, and 16.8% had a postgraduate degree. Of the total sample, 4.0% and 6.2% reported having currently or having had an eating disorder diagnosis, respectively, while 0.7% of women reported being pregnant. Finally, participants' BMI, calculated from self-reported height and weight, ranged from 11.48 to 55.35 kg/m² ($M = 27.56$, $SD = 6.44$).

2.2. Measures

2.2.1. Participant information

Participants were asked to self-report their age, gender identity, highest education degree, height (in foot and inches or meters), and weight (in pounds or kilograms). Additionally, participants were asked whether: (a) they were currently diagnosed with an eating disorder (0 = no; 1 = yes); (b) had been diagnosed with an eating disorder in the past (0 = no; 1 = yes); and, if they identified as a woman, (c) if they were pregnant. Data on eating disorder diagnosis and pregnancy were requested for sample descriptive purposes only.

2.2.2. Body acceptance by others

As recommended by Swami and Barron (2019), the BAOS-2 (Swami, Todd, Stieger et al., 2021) was translated from English into French as follows. First, the BAOS-2 items, instructions, and response scale were independently forward-translated by a professional translator and a member of the research team. Second, an independent translator (a member of the research team) examined these translations, resolved any discrepancies, and produced a single, synthesised French version. Third, the synthesised French version was back-translated into English by two independent professional translators. Finally, the forward- and back-translations were inspected by an expert committee comprising all previous translators and all members of the research team. Any minor discrepancies were resolved by discussion. The final version of the BAOS-2 in French are reported in the [Supplementary Materials](#). Responses to the items were made using a 5-point scale ranging from 1 (*never*; French: *jamais*) to 5 (*always*; French: *toujours*).

2.2.3. Body appreciation

Participants completed the Body Appreciation Scale-2 (BAS-2; Tylka

& Wood-Barcalow, 2015; French translation: Kertechian & Swami, 2017) to assess a core facet of positive body image. The 10-item BAS-2 assesses acceptance of one's body, respect and care for one's body, and protection of one's body from unrealistic beauty standards. Items were rated on a 5-point scale (1 = *never*, 5 = *always*) and higher scores reflect greater body appreciation. Scores on the BAS-2 have been shown to have a unidimensional factor structure, adequate composite reliability, and good construct validity, including in French-speaking Canadian adults (Swami, Tran et al., 2023). In the present study, McDonald's ω for scores on this scale was .958.

2.2.4. Dieting

Dieting-related attitudes and behaviours were assessed using the Dieting subscale of the Eating Attitudes Test-26 (EAT-26; Garner et al., 1982; French translation: Leichner et al., 1994), which includes 13 items that assess the degree of dieting-related attitudes and behaviours characteristic of disordered eating. All items on this subscale were rated on a 6-point scale ranging from 1 (*never*) to 6 (*always*), with higher scores reflective of greater dieting attitudes and behaviours. The French version of the EAT-26, including the Dieting subscale, has been shown to have adequate construct validity and composite reliability in French-speaking Canadian adults (Leichner et al., 1994). In the present study, McDonald's ω for scores on this subscale was .926.

2.2.5. Orthorexia

Participants completed the Teruel Orthorexia Scale (TOS; Barrada & Roncero, 2018; French translation: Maïano et al., 2022). This is a 17-item scale that assesses symptoms of orthorexia nervosa (i.e., an obsession or extreme preoccupation with eating healthily; 8 items) and what the authors refer to as “healthy” orthorexia (i.e., a non-pathological form of orthorexia; 9 items). All items were rated on a 4-point scale ranging from 0 (*completely disagree*) to 3 (*completely agree*), with higher scores reflective of greater orthorexic symptomatology. The French version of the TOS has been shown to have a 2-factor structure and adequate psychometric properties in French-speaking Canadian adults (Maïano et al., 2022). In the present study, McDonald's ω was .882 for the Orthorexia Nervosa subscale and .874 for the Healthy Orthorexia subscale.

2.2.6. Anxiety and depression

Participants completed the Hospital Anxiety and Depression Scale (HADS; Zigmond & Snaith, 1983; French translation: Savard et al., 1998) to assess anxiety and depression, respectively. The HADS consists of 14 items, seven of which assess symptoms of anxiety and seven of which assess symptoms of depression. Each item was rated on a 4-point scale, with total scores for each subscale ranging from 0 to 21. The French version of the HADS has been shown to have good sensitivity to determine the presence of anxiety or depressive symptoms, respectively (Savard et al., 1998). In the present study, McDonald's ω was .846 for symptoms of anxiety and .819 for symptoms of depression.

2.2.7. Life satisfaction

We asked participants to complete the Satisfaction With Life Scale (SWLS; Diener et al., 1985; French translation: Blais et al., 1989), a 5-item instrument that assesses an aspect of subjective well-being on a 7-point scale (1 = *strongly disagree*, 7 = *strongly agree*). Higher scores on this scale reflect greater life satisfaction. Scores on the French translation of the SWLS have been shown to have a unidimensional factor structure in French-speaking Canadian adults (Swami et al., 2025). In the present study, McDonald's ω for scores on this scale was .918.

2.2.8. Self-compassion

To measure self-compassion, we used the Short Form of the Self-Compassion Scale (SCS-SF; Raes et al., 2011; French translation: Swami et al., 2025), a 11-item scale that assesses multidimensional facets of self-compassion. All items were rated on a 5-point scale,

ranging from 1 (*almost never*) to 5 (*almost always*). Recent work has suggested that the SCS-SF is best conceptualised as consisting of two discrete subscales assessing compassionate (5 items) and uncompassionate self-responding (6 items), respectively, including in French-speaking Canadian adults (Swami et al., 2025). In the present study, we only used the compassionate self-responding subscale, for which McDonald's ω was .807.

2.2.9. Self-esteem

To assess self-esteem, we used the Single-Item Self-Esteem Scale (SISE; Robins et al., 2001; French translation: Swami et al., 2020). Participants were asked to rate the item "I have high self-esteem" on a 7-point scale, ranging from 1 (*not very true of me*) to 7 (*very true of me*), with higher scores reflective of higher self-esteem. Scores on the SISE have been reported to have adequate construct validity in diverse populations (Swami et al., 2020).

2.3. Procedure

The present study was carried out in accordance with the principles of the Declaration of Helsinki and authorisation was granted by the research ethics committee of the first and last authors' university (approval code: #2025–3656). The majority (95%) of participants was recruited in the Canadian province of Quebec via a panel from Leger Opinion and received CAD\$1.80 as remuneration. The remainder of participants (5%) were recruited in the community via social media advertising, targeted emails, advertising in a local newspaper, and private clinic websites. No compensation was offered for these participants. Data collection took place between November 2024 and April 2025. Participants provided digital informed consent and were asked to complete an anonymous survey hosted on LimeSurvey, in which the order of presentation of instruments described above was presented in a pre-determined order. All participants took part on a voluntary basis.

2.4. Statistical analyses

To accommodate an EFA-to-CFA strategy, the total dataset was split using a computer-generated random seed. The two split-half subsamples consisted of 382 and 381 participants, respectively. No significant differences were found between the two split-half subsamples in terms of age, $t(761) = 1.05$, $p = .292$, $d = 0.08$; BMI, $t(748) = 0.89$, $p = .373$, $d = 0.07$; gender identities, $\chi^2(5) = 4.39$, $p = .495$; and educational qualifications, $\chi^2(9) = 4.28$, $p = .892$. Therefore, any differences between the two split-half subsamples were assumed to have been minimised by the random-split procedure.

2.4.1. First split-half subsample

Mplus 8.11's (Muthén & Muthén, 2024) maximum likelihood robust (MLR) estimator was used to handle the missing data at the BAOS-2's item level (1.31%–10.73%, $M = 4.99\%$). Three exploratory factor analyses (EFAs) were performed on the BAOS-2's items using an oblique geomin rotation and an epsilon value of .5 (Marsh et al., 2009, 2014). The best EFA solution was retained based on the Horn's (1965) parallel analysis test and the following fit indices (Hu & Bentler, 1999; Marsh et al., 2005): the Comparative Fit Index (CFI), Tucker Lewis Index (TLI), and root mean square error of approximation (RMSEA). A model was considered as having an acceptable or good fit to the data when CFI and TLI values were equal to or higher than .90 or .95 and RMSEA values were lower than or equal to .08 or .06, respectively (Hu & Bentler, 1999; Marsh et al., 2005). Finally, the McDonald's (1970) ω was used to estimate the composite reliability of latent factor(s).

2.4.2. s split-half subsample

The best EFA solution retained from the first split-half subsample was examined using Mplus 8.11's (Muthén & Muthén, 2024) MLR estimator (missing at BAOS-2's item level was between 1.57% and 8.40%, $M =$

4.62%) and CFA. The same fit indices as reported above (first split-half subsample) were used to assess model fit.

2.4.3. First and second split-half subsamples

Using Mplus 8.11 (Muthén & Muthén, 2024) MLR estimator, we assessed invariance of the optimal solution across subsamples using a CFA model and with the following sequence (Millsap, 2011): (i) configural invariance; and invariance of (ii) loadings; (iii) intercepts; (iv) uniqueness; (v) latent variance; and (vi) latent mean factor. Comparison of the models (i.e., the subsequent model is compared to the previous one) were based on changes (Δ) in CFI, TLI, and RMSEA. When Δ CFI and/or Δ TLI decreases were higher than .01, and/or the increase in Δ RMSEA was higher than .015, a sequence was considered as non-invariant (Chen, 2007; Cheung & Rensvold, 2002).

A hybrid MIMIC multiple-group model (Maïano et al., 2023; Morin et al., 2018) was used to examine presence of DIF and latent mean differences as function of participants' age, BMI, and gender identity, as well as the equivalence of these associations across subsamples. This method was favoured relative to multi-group invariance testing, as we were examining multiple continuous predictors (i.e., age and BMI) that should not be arbitrarily categorised (Swami, Maïano et al., 2023). These models were built from the most invariant model identified during the measurement invariance test as function of subsamples. Hybrid MIMIC models were estimated in the following sequence (Morin, 2023; Swami, Maïano et al., 2023): (a) null effects model; (b) saturated model; and (c) factors only model. To facilitate interpretation, age and BMI were standardized prior to the analyses. These models were first examined with all associations freely estimated across subsamples, and then the most appropriate model retained was compared to a model in which all associations were constrained to be equal across subsamples.

First, we considered that there was a meaningful association between participants' age, BMI, and gender identity and the BAOS-2's item responses when the increase of Δ CFI and/or Δ TLI was higher or equal to .01 and/or the decrease of Δ RMSEA was higher or equal to .015 between the saturated and factors only models relative to the null effects model. Second, we considered that there was a DIF when an improvement in model fit is observed for the saturated model relative to the factor only model. Third, we considered that there was a subsamples equality in the association between participants' age, BMI, and gender identity and the BAOS-2's item responses when lack of improvement in model fit was observed between the most appropriate hybrid MIMIC model and an alternative version of this model in which all associations were constrained to be equal across subsamples.

Construct validity of the BAOS-2 was examined using a structural equation model (SEM) in which the BAOS-2 latent factor(s) of the best factor solution was correlated with the observed scores of measures of anxiety and depression, body appreciation, dieting behaviour, orthorexia nervosa and healthy orthorexia, life satisfaction, compassionate self-responding (self-compassion), and self-esteem. Moreover, observed scores were all correlated in this SEM.

3. Results

3.1. First split-half subsample

Fit indices for the three EFA models are presented in Table 1. All three factors solution had acceptable-to-good fit to the data (Models 1–1 to 1–3). Results from the parallel analysis test showed that the data were best represented by a single-factor solution (for a visual representation, see Figure S1 in the Supplementary Materials). Examination of the 2-factor solution showed that the first factor comprised 10 of the 13 items and that, in the secondary factor, one of the three items (Item # 11 – "I can count on important others to accept my body") provided improper standardised factor loading (1.13) and uniqueness (<.071). Finally, examination of the 3-factor solution revealed that the first factor was split into two factors and that the third factor only comprised a single item

Table 1
Goodness-of-fit statistics for the BAOS-2.

Models	Samples	N°	Description	R ²	df	CFI	TLI	RMSEA	RMSEA 90% CI		CM	ΔR^2	df	p	ΔCFI	ΔTLI	$\Delta RMSEA$
									LB	UB							
EFA	First-split-half	1-1	1 factor	160.071*	65	.946	.935	.062	.050	.074	-	-	-	-	-	-	-
		1-2	2 factors	117.862*	53	.963	.946	.057	.043	.070	-	-	-	-	-	-	-
		1-3	3 factors	75.985*	42	.981	.964	.046	.029	.062	-	-	-	-	-	-	-
CFA	Second-split-half	2-1	1-factor	195.949*	65	.937	.917	.073	.061	.085	-	-	-	-	-	-	-
		3-1	Configural invariance	355.249*	130	.938	.926	.067	.059	.076	-	-	-	-	-	-	-
		3-2	Loadings invariance	374.883*	142	.936	.930	.066	.058	.074	3-1	14.36	12	.28	-.002	+.004	-.001
		3-3	Intercepts invariance	391.539*	154	.935	.934	.064	.056	.071	3-2	6.97	12	.86	-.001	+.004	-.002
		3-4	Uniqueness invariance	388.288*	167	.939	.943	.059	.051	.067	3-3	12.46	13	.49	+.004	+.009	-.005
		3-5	Variance invariance	389.065*	168	.939	.944	.059	.051	.066	3-4	0.10	1	.75	.000	+.001	.000
DIF: Age, body mass index, and gender identity	First and second split-half	3-6	Latent mean invariance	391.419*	169	.939	.944	.059	.051	.066	3-5	2.38	1	.12	.000	.000	.000
		4-1	Null effects	590.155*	247	.922	.926	.061	.055	.068	-	-	-	-	-	-	-
		4-2	Saturated	361.605*	169	.956	.939	.055	.048	.063	4-1	255.38	78	< .001	+.034	+.013	-.006
		4-3	Factors only	539.611*	241	.932	.934	.058	.051	.064	4-1	66.22	6	< .001	+.010	+.008	-.003
		4-4	Partial factors only	452.928*	224	.948	.945	.052	.045	.059	4-2	78.36	55	< .001	-.008	+.006	-.003
		4-5	Partial factors only (invariant across samples)	455.145*	227	.948	.946	.052	.045	.059	4-4	0.29	3	.96	.000	+.001	.000

Note. BAOS-2 = Body Acceptance by Others Scale-2; R² = robust maximum likelihood chi-square; df = degrees of freedom; CFI = comparative fit index; TLI = Tucker-Lewis index; RMSEA = root mean square error of approximation; 90% CI = 90% confidence interval of the RMSEA; LB = lower bound; UB = upper bound; Δ = change from the previous model; ΔR^2 = robust chi-square difference tests (calculated from loglikelihoods for greater precision); EFA = exploratory factor analyses; CFA = confirmatory factor analyses; DIF = differential item functioning. * $p \leq .05$

(Item # 11) with improper standardised factor loading (1.86) and uniqueness (-2.40). Therefore, based on the combination of fit indices, analyses of the factor solutions, and results of parallel analysis test, the unidimensional solution with all 13 items was retained for subsequent analyses. Table 2 presents the standardised parameters and the composite reliability coefficient for this model. Loadings were moderate-to-high ($\lambda = .573-.863$; $M_\lambda = .775$) and the composite reliability coefficient was excellent ($\omega = .952$).

3.2. Second Split-half subsample

Fit indices of the unidimensional CFA-based model are presented in Table 1. As can be seen, model fit was acceptable, loadings' values were moderate-to-high ($\lambda = .572-.876$; $M_\lambda = .772$; see Table 2), and the composite reliability coefficient of the BAOS-2's latent factor was excellent ($\omega = .951$).

3.3. First and second split-half subsamples

3.3.1. Subsample invariance

Fit statistics of the unidimensional CFA-based from tests of measurement invariance as function of subsamples are presented in Table 1 (Models 3-1 to 3-6). Results supported complete measurement invariance (i.e., configural, loadings, intercepts, uniqueness, variance, and latent mean) across subsamples.

3.3.2. Differential item functioning across both subsamples

The results from the hybrid MIMIC models are presented in Table 1. These analyses start with the latent mean invariance model (Model 3-6) that was the most invariant model across subsamples. Results showed that age, BMI, and gender identity were associated with responses on the BAOS-2. However, they also indicated DIF as a function of age, BMI, and gender identity given than the saturated model resulted in improvement in model fit relative to the factors only model ($\Delta CFI = +.024$, $\Delta TLI = +.005$, $\Delta RMSEA = -.003$). Inspection of the modification indices of the factors-only model revealed that DIF could be attributed to six paths from age to BAOS-2's items (i.e., Items #1, 6, and 8 for the first-split half subsample; Items #4, 10 and 13 for the second-split half subsample), five paths from BMI to BAOS-2's items (i.e., Items #4, 6, 11, and 13 for the first-split half subsample; Item #5 for the second-split half subsample), and six paths from gender identity to BAOS-2's items (i.e., Items #4, 5, and 10 for the first-split half subsample; Items #1, 6, and 13 for the second-split half subsample). Therefore, the constraints of these paths were relaxed in a partial factors-only model. Comparison of the

Table 2

Standardised parameter estimates from the exploratory and confirmatory factor analytic representation of the BAOS-2 in the first and second split-half subsamples.

Items	First split-half subsample		Second split-half subsample	
	λ	δ	λ	δ
1	.573	.671	.572	.673
2	.815	.336	.848	.281
3	.761	.421	.815	.336
4	.806	.350	.842	.291
5	.591	.651	.663	.560
6	.850	.278	.831	.309
7	.816	.335	.814	.337
8	.834	.304	.825	.320
9	.863	.255	.876	.233
10	.804	.354	.838	.298
11	.814	.337	.770	.406
12	.851	.277	.763	.418
13	.699	.511	.573	.671
ω	.952		.951	

Note. BAOS-2 = Body Acceptance by Others Scale-2; λ = factor loadings; δ = uniquenesses; ω = McDonald's omega.

partial factors-only model with the saturated model did not show a substantial decrease of fit indices. Finally, the model built from the partial factors-only model (Model 4-5) provided support for the invariance across subsamples of these associations between the BAOS-2's latent factor and age, BMI, and gender identity.

The results from the partial factors-only model invariant across subsamples are presented in Table 3. First, results showed a lack of significant association between the BAOS-2's latent factor and age and gender identity, respectively. However, the results also showed that BMI was significantly and negatively related to the BAOS-2's latent factor. Therefore, participants with higher BMIs tended to present significantly lower levels of body acceptance by others relative to participants with lower BMIs.

The SEM examining the construct validity of the unidimensional model of the BAOS-2 resulted in a good level of fit to the data, $\chi^2(173) = 531.377$, CFI = .951, TLI = .935, RMSEA = .052 (90% CI = .047; .057). As illustrated in Table 4, latent body acceptance by others was significantly and positively related to body appreciation, compassionate self-responding (self-compassion), life satisfaction, healthy orthorexia, and self-esteem. On the other hand, body acceptance by others was significantly and negatively related to orthorexia nervosa, anxiety, depression, and dieting.

4. Discussion

Overall, the results of the present study indicate that our novel French translation of the BAOS-2 retains its robust psychometric properties in a sample of French-Canadian adults. In terms of the factor structure of the French BAOS-2, the results of both EFA and CFA supported a unidimensional model with all 13 items retained. This model was fully invariant across both split-half subsamples. Yet, we also found that the BAOS-2 has differential item functioning based on respondent age, BMI and gender identity as function of subsamples. In both subsamples, higher respondent BMI was associated with lower body acceptance. Finally, we found strong evidence of convergent and concurrent validity for the French BAOS-2, with all associations with additional constructs in the expected direction.

In terms of factorial validity, EFA and parallel analysis supported the extraction of a unidimensional model of the French BAOS-2, with all 13 items retained. This model was cross-validated using CFA in a second split-half subsample, with the model showing adequate fit indices and additionally achieving full invariance across split-half subsamples. This suggests that scores on the French BAOS-2 are best conceptualised as unidimensional, which is consistent with the results of both the parent

Table 3
Relationships between the BAOS-2 latent factor and the predictors.

	b(SE)	Subsamples-specific standardised coefficients	
		β (First split-half subsample)	β (Second split-half subsample)
Age	.049 (.039)	.046	.046
Body mass index	-.325 (.044)*	-.309*	-.309*
Gender identity	.024 (.076)	.011	.011

Note. BAOS-2 = Body Acceptance by Others Scale-2; b = unstandardized regression coefficient taken from the partial Factors only invariant across samples (Model 4-5); SE = standard error of the coefficient; β = subsample-specific standardised regression coefficient (although some of the relations are invariant across subsamples, the standardised coefficients may still show some variation as a function of within-subsamples estimates of variability). Because age and body mass index were standardised prior to these analyses and that the BAOS-2 factor is estimated based on a model of latent variance invariant in which all latent factors have a SD of 1, all unstandardised coefficients can be directly interpreted as SD units. * $p \leq .001$.

Table 4
Construct validity of the BAOS-2 in the overall sample.

	BAOS-2
Body appreciation	.608*
Compassionate self-responding	.446*
Life satisfaction	.431*
Healthy orthorexia	.183*
Orthorexia nervosa	-.177*
Anxiety	-.294*
Depression	-.390*
Self-esteem	.415*
Dieting	-.376*

Note. BAOS-2 = Body Acceptance by Others Scale-2. * $p \leq .001$.

study utilising an English variant (Swami, Todd, Stieger et al., 2021), as well as translations of the BAOS-2 in other linguistic groups (Behrend et al., 2023; Geller et al., 2024; Modrzejewska et al., 2024; Nerini et al., 2024; Swami et al., 2021; Wang et al., 2022). Importantly, where some previous studies have reported deflated item-factor loadings for Item 13 (Modrzejewska et al., 2024; Swami et al., 2021), we found no evidence of this in the present work.

In a departure from the analytic strategies in previous studies, where examinations of measurement invariance have been limited to gender identity, we used hybrid MIMIC models to test for measurement biases as a joint function of age, and BMI, and gender identity. Our results indicated that all three variables were associated with item responses on the French BAOS-2, and inspection of paths indicated that across split-half subsamples some items intercepts were not invariant as function of age, and BMI, and gender identity. Relaxing the intercepts constraints of relevant items in a factors-only model allowed us to achieve a final model that was partially invariant across split-half subsamples. In this model, we found a lack of significant association between the BAOS-2 latent factor and age and gender identity, respectively. Broadly speaking, these results are consistent with the finding that the BAOS-2 achieves strong invariance across women and men, and that there are no meaningful gender differences in body acceptance by others (Behrend et al., 2023; Modrzejewska et al., 2024; Nerini et al., 2024; Swami et al., 2021; Wang et al., 2022). Additionally, where associations between body acceptance by others and age have been investigated, the association has been found to be very weak (Wang et al., 2022).

On the other hand, our results indicated that participant BMI was inversely associated with body acceptance by others. Previous studies have generally reported that the association between body acceptance by others and BMI is inverse, although the strength of association varies across studies, from being negligible (e.g., Nerini et al., 2024) to weak-to-moderate (Moelle & Papps, 2025; Swami et al., 2021; Swami, Todd, Stieger et al., 2021) and others reporting no significant association (Nerini et al., 2024). Swami, Todd, Stieger et al. (2021) have previously suggested that the negative association may reflect a cultural valuation of thinness that is increasingly found in most urbanised contexts (for a discussion, see Swami, 2015, 2021). Individuals who meet cultural prescriptions for thinness are, therefore, more likely to experience or perceive body acceptance by others. However, in keeping with recent discussions that have problematised the use of BMI (Bailey, Bessey, Lamarche, & Griffin, 2025), it is likely that BMI is only tapping a different underlying effect driven by weight bias internalisation. Consistent with this suggestion, Swami, Todd, Stieger et al. (2021) reported that associations between body acceptance by others and weight bias internalisation were stronger than associations with BMI.

Finally, the results of the present study also suggest that the French BAOS-2 has adequate construct validity. More specifically, convergent validity was upheld through a positive and moderate-to-strong association with body appreciation, while concurrent validity was supported through moderate associations with psychological well-being (anxiety, depression, life satisfaction, self-esteem), eating patterns (orthorexia, dieting behaviour), and self-compassion. Overall, these results are

consistent with previous findings indicating that greater body acceptance by others is associated with higher psychological well-being and lower symptoms of maladaptive eating (Avalos & Tylka, 2006; Swami, Todd, Stieger et al., 2021). However, the results also suggested that body acceptance by others was possibly more strongly associated with indices of well-being and self-compassion compared with negative eating patterns.

It may be that stronger associations with eating patterns would have been achieved had we included an index of adaptive eating. For instance, the Acceptance Model of Intuitive Eating (Avalos & Tylka, 2006) suggests that greater body acceptance by others is associated with more positive body image, which in turn is associated with adaptive eating styles, such as intuitive eating. As such, we may expect stronger associations between body acceptance by others and an index such as intuitive eating compared with orthorexia or dieting behaviours. This would be a useful avenue for future research. Certainly, our results currently only hint at the value of the Acceptance Model in the French-Canadian context, but this could be formally tested as the necessary instrumentation – including the French BAOS-2 – are validated for French-speaking adults.

The present results should be considered in light of several limitations of our work. First, because of recruitment method – via an online panel, supplemented by online community sampling – we caution that our sample of Québécois participants is unlikely to have been representative. This sampling strategy may have introduced biases that, in turn, could have affected the generalisability of our findings. Additionally, we did not gather potentially relevant information about urbanicity, socioeconomic status and food insecurity, racialised identity, and religion or religiosity of our sample. In a similar vein, while the objective of the present study was to validate a French translation of the BAOS-2, it may also be interesting to examine invariance of the latent body acceptance by others constructs in a sample of Francophone and Anglophone Canadians. Relatedly, it may also be important in the future to examine the extent to which adaptations are necessary to ensure that the BAOS-2 can also be completed by younger age groups. This is important in view of findings showing that appearance-based commentary is common in Québécois adolescents and have a negative impact on body image outcomes (Aimé & Maïano, 2014; Gagné et al., 2023). Finally, a small minority of participants reported having (currently or in the past) an eating disorder diagnosis or were pregnant at the time of study, and we do not know how these characteristics may have influenced our results.

Other limitations of the present study include the fact that we did not assess temporal stability or test-retest reliability of the French BAOS-2. These issues should be rectified in the future. Additionally, the assessment of the construct validity of the French BAOS-2 could also be extended through the inclusion of measures of pressure to conform to appearance ideals. For instance, previous work has shown that body acceptance by others is negatively associated with pressure to conform to appearance ideals from the media, peers, family, and significant others (Nerini et al., 2024). Likewise, it will be important in future work to assess the degree to which the French BAOS-2 evidences discriminant validity (e.g., through associations with impression management).

Additionally, given that the data were cross-sectional, directional relationships could not be established. Although unconditional body acceptance by others is hypothesised to promote positive body image in the Acceptance Model (Avalos & Tylka, 2006), some researchers instead found that body appreciation predicted increased unconditional acceptance by others, but not the reverse, among German women and men (Behrend et al., 2023). Therefore, it may be that those with positive body image are more likely to seek out body accepting environments, as suggested by recent qualitative research (Alleva et al., 2023), than positive body image being shaped by them. Clearly, assessing longitudinal relationships in future work within various geographic regions, including French-Canadian adults, is needed and would allow for tests of predictive validity and directionality (Wang et al., 2022).

Finally, our methodological choices may have affected data quality and validity. Given that the BAOS-2 is a self-report measure, we cannot rule out acquiescence and response biases, and given that the measures were administered in a pre-determined order without attention checks, we cannot rule out order effects and inattentive responding. In future work, it may also be useful to randomise the order of presentation of instruments within the survey to minimise order effects. Greater care could also have been taken to identify and screen out bots and those not paying full attention to the survey, although we note that these issues are typically minimised in panel responding.

Taken together, this study results indicate that our French translation of the BAOS-2 retains its strong psychometric properties, including a unidimensional factor structure and evidence of construct validity, in French-Canadian adults. We recommend two important next steps for researchers interested in the construct of body acceptance by others. For scholars in Quebec specifically, one important way to extend the current findings is to assess the fit of the Acceptance Model of Intuitive Eating – in which body acceptance by others is an important component – in Québécois populations. In a similar vein, the availability of the French BAOS-2 may be useful for practitioners with an interest in interventions that are focused on promoting body acceptance by others as a route toward more positive body image or healthier eating practices. For scholars internationally, it will be important to examine the extent to which latent body acceptance by others is invariant across national and linguistic contexts. This will help researchers better understand the ways in which cultural values and factors may shape body acceptance by others, which in turn has implications for the development and maintenance of positive body image.

CRediT authorship contribution statement

Christophe Maïano: Conceptualization, Methodology, Validation, Formal analysis, Data curation, Resources, Writing – original draft, Writing – review & editing, Software, Supervision, Visualization, Project administration, Investigation, Funding acquisition. **Viren Swami:** Conceptualization, Methodology, Validation, Writing – original draft, Writing – review & editing, Supervision. **Tracy L. Tylka:** Conceptualization, Methodology, Validation, Writing – review & editing. **Annie Aimé:** Conceptualization, Methodology, Validation, Writing – review & editing.

Ethical statement

This study was approved by the research ethics committee of the Université du Québec en Outaouais (#2025–3656) and conducted in accordance with the Declaration of Helsinki. Participants provided digital consent prior to responding to the online survey.

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Declaration of competing interest

The authors declare they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. Given her role as Editor-in-Chief of Body Image, Tracy L. Tylka had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.bodyim.2026.102045](https://doi.org/10.1016/j.bodyim.2026.102045).

Data availability

The authors do not have permission to share data.

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